

**SIMATS ENGINEERING**

**ASSESSMENT TOOL 3**

**COURSE CODE:MLA03**

**COURSE NAME: Reinforcement learning for Environment and Agent**

**COURSE OUTCOME COVERED**

**CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).**

**Assessment Tool Weightage: 25%**

**Dr G. A. SENTHIL**

**QUESTIONS: 10**

**Total Marks: 50**

**Annexure A**

**Data Interpretation**

**Duration: 1hr**

**Code of Conduct:**

**I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.**

**I understand that any violation of academic integrity rules will result in disciplinary action.**

**Signature of the student: K.Neeha**

### **1. Reinforcement Learning Reward Trend Analysis (5 Marks)**

#### **Interpretation**

A Reinforcement Learning (RL) agent is trained over multiple episodes, and its reward gradually increases from **10 to 45**. In RL, the reward is a numerical value that measures how well the agent performs after taking an action. A steady increase in reward indicates that the agent is learning from its interactions with the environment and improving its behavior over time.

Initially, the agent has little or no knowledge of the environment. It explores different actions through trial and error, which often results in low rewards. As the training continues, the agent gathers experience and identifies actions that lead to better outcomes. It updates its policy based on the rewards received and gradually learns to avoid poor decisions while selecting actions that maximize cumulative rewards.

The increasing reward trend also indicates that the learning algorithm is functioning correctly. Instead of repeatedly making random decisions, the agent starts exploiting previously learned knowledge. This improvement reflects better decision-making and efficient adaptation to the environment.

#### **Learning Performance**

- Initial reward = **10**, indicating limited knowledge and frequent exploration.
- Final reward = **45**, showing that the agent has learned effective actions.
- The gradual increase demonstrates continuous improvement in policy quality.
- The agent is successfully maximizing cumulative rewards over time.

### Policy Improvement

The policy determines which action the agent should take in each state. As rewards increase, the policy becomes more accurate because the agent chooses actions that provide higher long-term returns. This means the policy is moving closer to the **optimal policy**.

### Conclusion

The gradual increase in rewards from **10 to 45** demonstrates successful learning. It indicates that the RL agent is effectively improving its policy, learning from past experiences, and converging toward an optimal strategy that maximizes cumulative rewards. The reward trend is strong evidence of improved learning performance and better decision-making.

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## 2. Exploration vs Exploitation using $\epsilon$ -Greedy (5 Marks)

### Interpretation

An  $\epsilon$ -greedy agent is tested with  $\epsilon = 0.1, 0.3, \text{ and } 0.5$ , and the highest cumulative reward is obtained at  $\epsilon = 0.1$ , while the lowest reward is obtained at  $\epsilon = 0.5$ . This result illustrates the importance of balancing exploration and exploitation in Reinforcement Learning.

Exploration means trying new or less-visited actions to discover potentially better rewards, whereas exploitation means selecting the best-known action based on previous learning. Both are necessary because exploration helps the agent gain knowledge, while exploitation allows it to use that knowledge efficiently.

### Analysis

When  $\epsilon = 0.1$ , the agent exploits the best-known action **90%** of the time and explores only **10%** of the time. This allows the agent to make good decisions while still occasionally discovering better actions. As a result, it achieves the highest cumulative reward.

When  $\epsilon = 0.3$ , the agent explores more frequently. Although this may help discover new strategies, it also causes the agent to perform more random actions, leading to moderate rewards.

When  $\epsilon = 0.5$ , the agent explores half of the time. Excessive exploration reduces the opportunity to exploit the learned policy, resulting in lower cumulative rewards because many random actions are less effective.

### Comparison

#### Epsilon Exploration Exploitation Performance

|     |        |        |                 |
|-----|--------|--------|-----------------|
| 0.1 | Low    | High   | Highest Reward  |
| 0.3 | Medium | Medium | Moderate Reward |
| 0.5 | High   | Low    | Lowest Reward   |

## Conclusion

The results show that  $\epsilon = 0.1$  provides the best balance between exploration and exploitation. It enables the agent to maximize cumulative rewards while still maintaining enough exploration to improve its policy. Therefore,  $\epsilon = 0.1$  gives the best overall learning performance.

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## 3. Comparison of Q-Learning and SARSA (5 Marks)

### Interpretation

Q-Learning and SARSA are compared over **100, 200, and 300 episodes**, and Q-Learning consistently achieves slightly higher average rewards. This indicates that Q-Learning learns more efficiently and converges faster.

Q-Learning is an **off-policy** algorithm. It updates its action-value function using the maximum possible future reward, regardless of the action actually chosen. This aggressive learning strategy enables the agent to quickly identify the optimal policy.

SARSA is an **on-policy** algorithm. It updates its values using the action that the agent actually takes. Since it considers the current policy during learning, it behaves more cautiously and often converges more slowly.

### Analysis

The consistently higher rewards of Q-Learning indicate that it is better at identifying actions with higher long-term returns. SARSA, although more conservative, may avoid risky actions and therefore sometimes receives slightly lower rewards.

### Comparison

| Feature        | Q-Learning SARSA |                |
|----------------|------------------|----------------|
| Learning Type  | Off-Policy       | On-Policy      |
| Average Reward | Higher           | Slightly Lower |
| Convergence    | Faster           | Slower         |
| Risk Taking    | High             | Low            |
| Policy         | Optimal          | Safer Policy   |

### Conclusion

Based on the observed rewards, **Q-Learning converges faster** and produces higher average rewards than SARSA. Therefore, it is more effective when the objective is to maximize long-term cumulative rewards.

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## 4. Effect of Discount Factor ( $\gamma$ ) (5 Marks)

### Interpretation

The average reward increases as the discount factor changes from **0.2 to 0.5 to 0.9**. This indicates that considering future rewards improves the quality of decision-making.

The discount factor determines the importance given to future rewards. A low discount factor emphasizes immediate rewards, while a high discount factor encourages the agent to consider long-term benefits.

**Analysis**

With  $\gamma = 0.2$ , the agent focuses mainly on immediate rewards and ignores future opportunities. This often leads to short-sighted decisions.

With  $\gamma = 0.5$ , the agent balances immediate and future rewards, improving learning performance.

With  $\gamma = 0.9$ , the agent strongly values future rewards. It selects actions that may not produce immediate benefits but lead to higher cumulative rewards in the long run. This results in the highest average reward.

**Comparison**

| Discount Factor | Decision Focus   | Average Reward |
|-----------------|------------------|----------------|
| 0.2             | Immediate Reward | Low            |
| 0.5             | Balanced Reward  | Medium         |
| 0.9             | Long-Term Reward | High           |

**Conclusion**

The experimental results show that  $\gamma = 0.9$  is the most suitable discount factor because it allows the agent to maximize long-term cumulative rewards. It leads to better policy learning, improved convergence, and higher overall performance.

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**5. Success Rate Analysis (5 Marks)**

**Interpretation**

The success rate of the RL agent increases from **45% after 50 episodes** to **92% after 200 episodes**. This significant improvement indicates that the agent is learning successfully and making increasingly effective decisions.

At the beginning of training, the agent has very little experience. It explores many different actions, some of which are unsuccessful, resulting in a lower success rate. As more episodes are completed, the agent gains experience, improves its policy, and chooses better actions more consistently.

**Analysis**

The increase from **45% to 92%** demonstrates that the agent is learning from previous experiences. The policy becomes more accurate because the agent receives continuous feedback through rewards and penalties. Higher success rates also indicate reduced errors and better adaptation to the environment.

**Performance Summary**

**Episodes Success Rate Interpretation**

|     |      |                        |
|-----|------|------------------------|
| 50  | 45%  | Initial Learning Stage |
| 100 | ~70% | Policy Improving       |

### Episodes Success Rate Interpretation

|     |     |                     |
|-----|-----|---------------------|
| 200 | 92% | Near Optimal Policy |
|-----|-----|---------------------|

### Conclusion

The steady increase in success rate from **45% to 92%** clearly demonstrates that the RL agent has learned an effective policy. The data indicates strong learning performance, improved decision-making, faster convergence, and behavior that is very close to the optimal policy.